

FastAPI Backend Interview Notes & Questions

1. FastAPI Basics

- FastAPI is a modern Python web framework used to build APIs.
- It is fast because it supports asynchronous programming using asyncio.
- Common HTTP methods: GET, POST, PUT, DELETE.
- Path operation decorators: `@app.get()`, `@app.post()` etc.

Interview Questions
What is FastAPI?
Difference between GET and POST?
Why is FastAPI fast?

2. Asyncio & Async Programming

- `async def` creates asynchronous functions.
- `await` pauses the current task without blocking the server.
- asyncio event loop manages multiple waiting tasks.
- Useful for database queries, APIs, and network operations.

Interview Questions
Difference between sync and async programming?
What does <code>await</code> do?
What is an event loop?

3. JWT Authentication

- JWT stands for JSON Web Token.
- JWT has 3 parts: header.payload.signature.
- Used for authentication and protected routes.
- Token usually sent in Authorization header.

Interview Questions
What is JWT?
Authentication vs Authorization?
What is access token?

4. FastAPI Dependency Injection

- Depends() is used for dependency injection.
- Reusable logic like DB connection or token verification can be injected.

Interview Questions
What is Depends() in FastAPI?
Why dependency injection is useful?

5. Database Basics

- SQLAlchemy is commonly used with FastAPI.
- Session is used to interact with the database.
- CRUD = Create, Read, Update, Delete.

Interview Questions
What is ORM?
What is SQLAlchemy?
Explain CRUD operations.

6. Common Interview Tips

- Understand concepts instead of memorizing syntax.
- Practice debugging errors.
- Build small projects like Todo API, Login System, Blog API.

Interview Questions
Explain a project you built.
How do you debug backend errors?
What challenges did you face while learning backend?